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Assessing pollen nutrient content: a unifying approach for the study of bee nutritional ecology

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Poor nutrition and landscape changes are regularly cited as key factors causing the decline of wild and managed bee populations. However, what constitutes 'poor nutrition' for bees currently is inadequately defined. Bees collect and eat pollen: it is their only solid food source and it provides a broad suite of required macro- and micronutrients. Bees are also generalist foragers and thus the different pollen types they collect and eat can be highly nutritionally variable. Therefore, characterizing the multidimensional nutrient content of different pollen types is needed to fully understand pollen as a nutritional resource. Unfortunately, the use of different analytical approaches to assess pollen nutrient content has complicated between-studies comparisons and blurred our understanding of pollen nutrient content. In the current study, we start by reviewing the common methods used to estimate protein and lipids found in pollen. Next, using monofloral *Brassica* and *Rosa* pollen, we experimentally reveal biases in results using these methods. Finally, we use our collective data to propose a unifying approach for analysing pollen nutrient content. This will help researchers better study and understand the nutritional ecology—including foraging behaviour, nutrient regulation and health—of bees and other pollen feeders.

This article is part of the theme issue 'Natural processes influencing pollinator health: from chemistry to landscapes'.

1. Introduction

Recent reports of global declines in wild bee abundance and diversity [1] and managed honeybees [2] are well documented. Poor nutrition and landscape changes are two key factors often cited as causing these declines [3]. Thus, enhancing the landscape by increasing the availability of nutritious foraging resources is a top research priority for improving bee demographics and health [4]. Like all animals, bees require a broad suite of nutrients, and consumption of these nutrients in a balanced manner maximizes survival, growth and fitness [5]. Bees collect and eat pollen, which is their primary source of nutrients, including amino acids and lipids, plus a range of vitamins, minerals and other micronutrients [6]. Pollen also contains carbohydrates, including monosaccharides (sugars) and polysaccharides (starch and cellulose). Monosaccharides, such as glucose and fructose, plus starch, can be used towards the energy needed for biosynthesis of tissues, proteins and enzymes [7], while polysaccharides provide resistance to pollen desiccation and structural support [8]. Importantly, some pollen is more effective than others in terms of improving bee performance and mitigating health problems [9], likely because the nutrient content of different floral species varies widely [10,11]. Thus, characterizing and assessing the multidimensional nutrient content of pollen is needed to inform decision-making processes related to the selection of bee-attracting plants for land management, as well as for the

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creation of artificial pollen substitutes for supplementary feeding of managed bees [12].

Characterizing the multidimensional nutrient content of pollen is often not straightforward for two main reasons. First, a determination needs to be made with respect to what nutrients to measure. Traditionally, studies have emphasized quantifying total macronutrient pools (proteins, carbohydrates and lipids), and there are a range of methods that can be used for each of these broad macronutrient classes. Proteins provide the bulk of the amino acids that bees require to build the protein they need: (1) for structural purposes, (2) as enzymes, (3) for transport and storage, and (4) as receptor molecules [13]. Protein amounts can be measured directly using colorimetric measures (e.g. Bradford, bicinchoninic acid (BCA) and Lowry assays) or estimated by measuring nitrogen (N) content (e.g. Dumas and Kjeldahl assays). Carbohydrates exist as two main pools in pollen—structural and non-structural—but only non-structural carbohydrates (e.g. simple sugars and starch) are digestible and available for use in bees. Thus, carbohydrate analysis of pollen is typically restricted to measuring digestible carbohydrates [14]. It is also worth noting that some bees add nectar to the pollen they collect; this binds the pollen together during collection and for transportation back to the hive. The lipid pool of pollen is very diverse and includes fatty acids, triacylglycerols, sphingolipids, galactolipids, glycerophospholipids and sterols [15]. Bees can synthesize many fatty acids and phospholipids [13] but require a dietary source of two polyunsaturated fatty acids (PUFA)—linoleic acid and α-linolenic acid [16]—and sterols [17]. However, the total fatty acid pool and total sterol pool compose only a small fraction of the total lipid pool in pollen. As a whole, total lipid content can be measured using gravimetric methods (Folch, Bligh & Dyer and Loveridge assays) and colorimetric methods (e.g. vanillin assay) methods. Identifying and quantifying the total fatty acids and sterols in pollens typically employs gas chromatography–mass spectrometry (GC-MS) techniques.

Bees also require a suite of micronutrients that exist as molecular organic compounds (e.g. vitamins, enzymes and co-enzymes), positively charged cations (e.g. Na^+, K^+, Ca^{2+}, Mg^{2+}, Fe^{2+}, Zn^{2+} and Mn^{2+}), or negatively charged anions (PO_4^{3-}, Cl^-). One useful way to approach micronutrients is to focus on biologically important atomic elements [18]. Carbon (C), hydrogen (H), oxygen (O) and nitrogen (N) collectively account for *ca* 99% of the mass of most biological systems [19]; they are the dominant elements in organic biomolecules including proteins, carbohydrates and lipids. Phosphorus (P) is also an important ubiquitous element given its role in lipid membranes, ATP and nucleic acids [20]. The remaining essential elements of an organism's ionome [21] involved in animal physiological function are (in rough order of biomass concentration): potassium (K), sulfur (S), calcium (Ca), magnesium (Mg), sodium (Na), iron (Fe), zinc (Zn), copper (Cu) and manganese (Mn). Kaspari [18] split these nine elements into two groups: (1) structural micronutrients, and (2) fluxing micronutrients. Structural micronutrients (S, Zn, Mn, Fe, Cu) contribute to the shape and function of biomolecules. Fluxing micronutrients (Na^+, K^+, Ca^{2+}, Mg^{2+}) serve osmoregulatory, nervous and motor systems; they move back and forth across membranes. Elemental analysis is often performed using a nitric acid and plant digest method paired with inductively coupled plasma spectrometry.

The second challenge related to assessing pollen nutrient content is that there are inconsistencies with how researchers process pollen. Mature pollen has a three-layer outer wall comprising the: (1) pollen coat, (2) exine and (3) intine [22]. The pollen coat (a.k.a. 'pollenkitt') is the hydrophobic outermost surface of the pollen grain and is composed of lipids, proteins, pigments and other compounds, including carotenoids. A lipid-rich adhesive material on the pollen coat helps facilitate pollination [23–25]. The exine is the second layer. It is primarily made up of sporopollenin, a lipid- and phenolic-based polymer that is highly resistant to physical and chemical stress. The third layer, the intine, is the inner structure of the pollen wall and it is composed of celluloses, pectin and proteins. The internal domain, or cytoplasm, contains proteins, free amino acids, polar and neutral lipids (e.g. triacylglycerols, or TAGs) and vitamins, most of which are water-soluble [15]. Additionally, as in plant leaf tissue, there is a suite of free ions. The cytoplasm can be particularly rich in lipids. In *Brassica* pollen, for example, approximately two-thirds of the lipids are contained within the internal domain [26]. The purpose of the pollen's three-layered outer wall is to help facilitate pollination and to protect the male gametophyte from environmental variables, including desiccation and UV radiation [25,27]. The tough nature of pollen has driven animals to evolve unique mechanisms to rupture the pollen wall to extract nutrients [11]. Because of its tough and sometimes impenetrable nature, standard methods to extract nutrients from pollen grains may not be enough to capture the internal components of pollen [28,29]. Some, but not all, researchers disrupt the pollenkitt prior to analysis. Most recently, Campos *et al*. [30] noted the need to disrupt the pollen wall to determine the phenolic compounds in pollen. In theory, this step increases extraction success because an intact pollenkitt may prevent solvents from entering the centre of the pollen grain [29]. For example, Somerville [31] and Evans *et al*. [28] used different methods when measuring the total lipid content of *Brassica napus* pollen and reported very different results. Somerville [31] used diethyl ether as the solvent to extract intact pollen grains and reported a total lipid value of 7.1%. In contrast, Evans *et al*. [28] used a chloroform/methanol treatment on disrupted pollen grains and reported a total lipid value of 25.4%.

In the current paper, we start with a brief review. It covers pollen structure and how it might influence nutrient extraction assays, and compares the key analyses used to measure total protein and total lipid content in pollen. We focused on protein and lipid analyses because they are the two macronutrients in pollen that are most often linked to bee health. They are also the nutrients with the highest amount of variation in reported results. In the next section, we use bee-collected pollen from plants belonging to the genera *Brassica* and *Rosa* to analyse: (1) total protein content, (2) total lipid content, (3) total fatty acid content and composition, (4) total digestible carbohydrate content, and (5) amount of ten biologically significant elements (P, K, S, Ca, Mg, Na, Zn, Fe, Mn and Cu). Importantly, we experimentally compare different protein and lipid analytical techniques, including on intact and disrupted pollen. Based on our findings, which acknowledge the unique properties of pollen, we make unifying recommendations for analysing pollen for nutrition studies. This will help researchers better study and understand the nutritional ecology (foraging behaviour, nutrient regulation and health) of bees and other pollen feeders.

2. Review of methods for estimating pollen nutrient content

The growing need for pollination services, coupled with the need to protect wild bee populations, has stimulated research focused on generating a more comprehensive understanding of bee nutrition [32]. For bees, dietary carbohydrates primarily originate from nectar but some are present in pollen. By contrast, pollen constitutes the sole source of dietary protein and lipids. There are multiple methods available for quantifying protein and lipid content, and while most methods are sound in the context of the study they were used for, between-studies comparisons that used different methods are often problematic and hinder the study of pollen-eating insects on the global scale. In this section, we review the protein and lipids assays commonly used for the examination of pollen nutrient content. We briefly describe how each is performed, noting their advantages and limitations.

(a) Pollen structure and its potential impact on pollen nutrient analysis

The pollen grain protects the male gametophyte and facilitates transmission to the female reproductive plant structures for fertilization. In angiosperms, pollination is largely facilitated by animals moving between flowers, collecting or feeding on pollen grains. This mutualism led to the evolution of different structures that together promote successful pollination: a three-layered outer wall and a nutritionally rich inner domain containing the male gametophyte. While a great deal of structural variation exists between pollen grains, including the number and types of apertures (openings), the general plan of organization of a pollen grain is conserved across taxa (reviewed by Campos *et al*. [30]). As a result, the toughness and porous nature of pollen walls can vary greatly between pollen taxa [33]. This variation can impact the efficiency with which nutrients are extracted, for both animals [11,34] and humans, in the context of pollen nutritional analyses [28,29]. Thus, variation in pollenkitt toughness, as well as aperture type and number, likely affects the permeability to solvents used to extract nutrients from pollen, which in turn may contribute significantly to variation in estimates of pollen nutrient content. One way to minimize this variation is to disrupt pollen grains mechanically prior to performing non-destructive nutrient analyses (e.g. Bradford, Lowry, Folch, vanillin assays). Unfortunately, many pollen nutritional studies fail to disrupt the pollen coat prior to analyses. This may result in an underestimation of pollen nutrient content, especially when using non-destructive sampling techniques.

(b) Estimating pollen protein content

The five most common methods used for the quantification of pollen protein are the Kjeldahl, Dumas, Bradford, Lowry and bicinchoninic acid (BCA) assays [35]. These methods indirectly estimate protein content based on the measure of one protein component and can be grouped based on their general principle: (1) the Kjeldahl and the Dumas assays measure nitrogen content; (2) the Bradford assay measures basic amino acid residues content, and (3) the Lowry and BCA assays measure the quantity of peptide bonds. After each assay has been performed, protein content is estimated using a relevant conversion factor (Kjeldahl and Dumas) or protein standard (Bradford, Lowry and BCA).

(i) Protein estimates using nitrogen content

Dumas combustion assay—In this method, the entire sample undergoes combustion under pure oxygen. This produces di-nitrogen, nitrogen oxides, carbon dioxide and water. Nitrogen oxides are then reduced into di-nitrogen with copper. Finally, nitrogen is quantified using a thermal conductivity detector.

Kjeldahl digestion assay—In this method, concentrated sulfuric acid is used to digest nitrogen into ammonium sulfate at high temperature. The ammonium sulfate is then distilled with sodium hydroxide to produce ammonia, which is introduced into a standard acid. The excess acid is then titrated to estimate ammonia content, which in turn estimates nitrogen content.

Conversion of nitrogen values to protein content—For both methods, the amount of nitrogen measured is converted to protein content. Because the amount of nitrogen varies across amino acids (from approximately 13 to 19%), and amino acid composition varies between proteins, the accurate determination of the conversion factor depends on the amino acid composition of the sample. For instance, standard conversion factors range from 5.18 for almonds to 6.38 for cow milk [36], with a most commonly used conversion factor of 6.25, based on the average amount of nitrogen (16%) in amino acids.

Advantages and limitations—These two methods generally produce very similar nitrogen values, with the Kjeldahl method producing slightly higher values than the Dumas method [37]. In the context of pollen protein analysis, the main difference is the minimum nitrogen detection limit, which depends on the amount of pollen required per sample. Compared with Kjeldahl, the Dumas limit is 100 times lower. Thus, the Dumas method accommodates smaller samples, which is an advantage considering that pollen collection, especially by hand, can be tedious and time consuming. There are, however, two main limitations common to both the Dumas and Kjeldahl assays. First, both methods assume that all the nitrogen contained in pollen originates from proteins. However, samples can contain sources of non-proteinaceous nitrogen in various proportions, including nucleic acids, amines, phospholipids and nitrogenous glycosides [38], even if these proportions remain relatively small in comparison with protein content [39]. Second, they both require the determination of a nitrogen conversion factor for each pollen species, or assume that pollen proteomes are similar enough across species to apply a single conversion factor. Multiple authors have proposed conversions factors for pollen, ranging from 3.8 to 5.6 [35,39–41]. Despite these different suggestions for lowering the nitrogen conversion factor for pollen, most studies still use 6.25 as the standard for pollen protein content [35].

(ii) Protein estimates measuring basic amino acid residues

Bradford assay—This colorimetric method uses Coomassie brilliant blue G250 dye, which shifts colour when binding with soluble proteins. The dye first forms a complex with protonated amino acid residues that changes the protein conformation, exposing its hydrophobic regions. The dye then binds to these regions, shifting its colour from red to blue [14,42–44].

Advantages and limitations—A strong advantage of the Bradford assay is that it requires very small amounts of material, as low as one-hundredth the amount of pollen

required for the Dumas combustion assay [35]. However, this assay has three main limitations when working with pollen. First, the assay only measures soluble proteins. The Coomassie brilliant blue G-250 dye has specific binding requirements and does not react with membrane proteins, small polypeptides, free amino acids or other nitrogen-containing material [44]. Second, it mainly has affinity for basic and aromatic amino acid residues, with a strong preference for arginine and lysine. Third, the assay is prone to interference from other components of pollen, such as sugars and lipids [45–47]. Modifications to the assay, such as changing the pH, adding NaOH or using different dyes, can correct variation, but doing so can increase opportunities for chemical interference [46].

(iii) Protein estimates measuring peptide bonds

The BCA and the Lowry assays are both colorimetric variants of the Biuret test, which estimates protein content by detecting peptide bonds. In the Biuret test, cupric ions in an alkaline solution are reduced to cuprous ions when binding to the nitrogen of peptide bonds, shifting the solution colour from blue to purple. These cuprous ions then react with the reagents of the BCA and Lowry assays.

BCA assay—In this method, each cuprous ion formed during the Biuret reaction is chelated with two molecules of bicinchoninic acid, producing a purple complex.

Lowry assay—In this method, the cuprous ions are oxidized back to cupric ions by the Folin–Ciocalteu reagent. This reaction involves the aromatic amino acid residues of tryptophan, and to a lesser extent tyrosine and cysteine, which lead to the production of a blue solution.

Advantages and limitations—Because the BCA assay directly measures the number of peptide bonds, it accurately estimates the total protein content. However, a pitfall of the BCA assay is the possible interference and overestimation of protein content due to lipids present in the sample [48]. This may be a problem given the high quantities of lipids in some pollen. Similar to the Bradford assay, the Lowry assay is specific to several amino acids and can, therefore, misestimate protein concentration in samples where these amino acids are over- or under-represented. It is also prone to overestimating protein content because the Folin–Ciocalteu reagent is sensitive to a suite of non-proteinaceous compounds [49].

(iv) Comparison of assays for the study of pollen protein content

Interference—Although Roulston *et al*. [35] did not find significant differences between the Bradford assay and nitrogen-based methods for determining protein content, there is significant variation among studies analysing the same pollen taxa using different approaches [11,28]. Spectrophotometric methods are prone to interference by various substances, which can yield inconsistent results [50]. For example, detergents, lipids and the combination of proteins and sugars can interfere with an assay to overestimate the protein content in a sample [45–47]. Thus, these interfering substances need to be minimized or eliminated to ensure accurate results [43]. Interference is not an issue for combustion assays (e.g. Dumas) that completely destroy the sample during analysis.

Pollen preparation—In the Bradford, BCA and Lowry assays, the pollen samples need to be properly prepared before being processed. Breaking open the pollen grains by mechanically or chemically disrupting the pollenkitt prior to analysis is a critical part of accurate protein determination. Using the Bradford assay, Evans *et al*. [28] detected significantly more protein in samples of disrupted pollen grains compared with intact pollen grains. In addition, different pollen taxa vary in how resistant they are to disruption [51]; the same disruption method might not be equally efficient with different pollen types. By contrast, nitrogen-based methods (e.g. Kjeldahl, Dumas) are not influenced by the pollen wall since the entire sample is destroyed during the assay.

(c) Estimating pollen lipid content

Although fewer studies have examined lipid content compared with protein, a range of methods can be used to quantify total lipid amounts in pollen. These approaches fall into two main categories: gravimetric and colorimetric. Gravimetric methods are based on the extraction of the lipid fraction of a sample and on the weighing of either the lipid fraction or the remaining, non-lipidic fraction of the sample. Colorimetric methods require the addition of a dye that binds to the lipid fraction of the sample and then quantification is done by measuring colour absorbance. It is also possible to go beyond those approaches to characterize the composition of lipid samples by chromatography and/or spectroscopy.

(i) Lipid estimates using gravimetric methods

Folch and Bligh & Dyer assays—The Folch [52] and the Bligh & Dyer assays [53] rely on the extraction of lipids using a combination of chloroform, methanol and water, in different ratios [54]. While the combination of chloroform and methanol is responsible for the lipid extraction, the addition of water creates a phase separation that washes the lipid phase from non-lipid contaminants [52,55]. Upon extraction, the lipid phase is weighed and compared with the dry weight of the sample to determine its lipid content.

Loveridge assay—The Loveridge assay [56] is more commonly used with whole body organisms [56,57]. It is based on the gravimetric measurement of the sample prior to the lipid extraction, and then again after extraction.

Advantages and limitations—Although these three methods are generally reliable, they suffer from two possible shortcomings. First, because small sample size increases variability, accurate measures require high-precision analytical balances. Second, the presence of extracted non-lipid contaminant material can overestimate lipid content [58,59]. This is directly linked to the efficacy of the solvent-based lipid extraction involved in these methods (reviewed in [60]). Lastly, lipids containing nonpolar groups, such as triacylglycerols, are readily extracted by nonpolar solvents such as hexane, benzene, chloroform or diethyl ether. By contrast, polar lipids are more efficiently extracted by solvents like methanol, which lie on the other end of the polarity spectrum. Consequently, an accurate lipid extraction of pollen samples calls for the knowledge of its lipid constituents, which can vary between pollen types.

(ii) Lipid estimates using colorimetric methods

Vanillin assay—The most common colorimetric approach for measuring lipid content is the sulfo–phospho–vanillin assay (SPV assay). The assay is based on an initial reaction between

sulfuric acid and fatty acid double bonds, followed by a reaction between the derived products and phosphoric acid [61].

Advantages and limitations—The two advantages of the SPV assay compared with gravimetric methods are: (1) the ability to use small samples, and (2) the fact that they are not impacted by the lipid extraction method. However, a significant disadvantage is that the SPV reagent only reacts with unsaturated fatty acids [62]. Since the initial reaction involves sulfuric acid and fatty acid double bonds, saturated fatty acids are not able to react with the SPV reagent [63,64]. Some studies have tried to modify the assay to measure total lipids in their specific samples of interest [65,66]. For example, the selection of a suitable oil to use as a standard was needed to estimate total lipids for a specific algal strain as a function of its fatty acid content [65].

3. Material and methods

(a) Pollen preparation

Pollen identification—We visually verified the identity and monofloral nature of the honeybee-collected, gamma-irradiated, *Brassica* sp. and *Rosa* sp. pollen samples used in our study. Bees commonly collect pollen types representing these two plant taxonomic groups. Each respective pollen type represented greater than 90% of the pollen grains in a sample.

Pollen drying—Pollen was stored at −20°C to prevent nutrient degradation [67,68]. Prior to each analysis (described below) all samples were freeze dried for 24 h (Labconco FreeZone freeze dryer). Freeze drying the pollen prevented lipids from binding to proteins and carbohydrates, which can occur when using high temperatures for drying [69].

Pollen disruption—To determine the effect of pollen disruption on protein and lipid analysis we compared non-disrupted and disrupted pollen. The disruption method for protein content analysis comprised three phases: (1) delipidation of the pollen exine with acetone to weaken the exine and facilitate further water absorption by osmosis; (2) rehydration of the pollen grains to increase internal water turgidity and help with the disruption process, and (3) mechanical disruption of the pollen with a hand-held tissue homogenizer. Additionally, full quantification of a sample's lipid content was done using an acetone wash to remove the lipids atop the pollenkitt, followed by physical disruption to release internal lipids within pollen.

To disrupt the pollen grains, we used a homogenizer in short sessions of 30 s; this prevented an increase in temperature due to friction of the pestle against the inner walls of the vials. We followed a standardized protocol to determine how many 30 s sessions were sufficient to disrupt all pollen grains: (1) 75 mg of dried and acetone-washed pollen was placed inside a 1.5 ml microcentrifuge tube; (2) 500 µl of deionized water was added to the vial; (3) the tip of a plastic pestle was introduced at the bottom of each vial; (4) the homogenizer was set at 3000 r.p.m. and homogenization took place for 30 s; (5) the pestle was taken out of the vial and both the vial and pestle were left to cool for 30 s; (6) during the cooling down period, 3 µl of the sample was collected and dispensed on a microscope glass slide; (7) steps 3 to 6 were repeated nine times, giving us a total of ten samples; (8) the ten slides were photographed at high resolution with a Keyence VHX-6000 microscope; (9) on each slide, three 1 mm^2 squares were randomly selected and the number of intact/disrupted pollen grains was counted. We scored a pollen grain as disrupted if its original shape was damaged/broken. At times, what seemed to be part of the cytoplasm was visibly protruding out of the ruptured grain. The randomly selected squares were counted by two different individuals and the values were averaged. The full details of our pollenkitt disruption protocol are provided in electronic supplementary material, S1.

(b) Protein analysis

As outlined above, choosing a protein assay can be challenging because each method has advantages and limitations [70]. Here, we compared two common protein analysis procedures, the Bradford and Dumas methods, to determine whether, and to what extent, they differed in their approximation of a pollen sample's total protein content. We also tested the influence of the pollen disruption process on the soluble protein fraction assays. The aim of this section is to provide quantifiable evidence of the variability observed when measuring protein concentration using different protein quantification methods.

Bradford assay—The Bradford assay was used to measure the water-soluble protein content (most of the protein found in plants) of *Brassica* and *Rosa* pollen. We performed a Bradford assay on non-disrupted pollen ($n = 3$), standardized with IgG from bovine serum albumin (BSA), following the protocol established by Vaudo *et al*. [71] and Deans *et al*. [14] (see electronic supplementary material for more detail). Before we conducted the Bradford assay, a protein extraction and solubilization step was required. This step was critical, as it determined how much protein was going to be available for the assay. We also performed the Bradford assay using disrupted pollen ($n = 5$ samples), which included extracting the lipid fraction prior to protein analysis. This step was added to make the nutrient contents within the pollen grain more accessible and to compare how pollen disruption affected the total protein yield. The full details of our Bradford assay protocol are provided in electronic supplementary material, S2.

Dumas assay—Samples ($n = 10$) of non-disrupted *Brassica* and *Rosa* pollen were sent to the Texas A&M AgriLife Extension Service's Soil, Water and Forage Testing Laboratory (soiltesting.tamu.edu) to determine the pollen's nitrogen content via the Dumas method, employing an Elementar vario MAX CN high-temperature analyser set at 950°C [72]. This assay depends on a high-temperature combustion process to determine the total nitrogen content of a sample. The protein content within each pollen sample was estimated using a 6.25 conversion factor. The full details of our Dumas assay protocol are provided in electronic supplementary material, S3.

(c) Lipid analyses

We used three different methods to determine the lipid content in *Brassica* and *Rosa* pollen: (1) the Folch assay, (2) the Loveridge assay, and (3) the vanillin assay. For all three methods, we used homogenized pollen samples that were split into two liquid fractions. One contained the purified lipids, while the other contained non-lipid molecules. Additionally, we identified and quantified the fatty acid profiles of samples that were processed through the Folch assay (details below). The full details of our Folch and Loveridge lipid assay protocols are provided in electronic supplementary material, S4. Protocols for the vanillin assay can be found in electronic supplementary material, S5.

Folch assay—Lipids were extracted from 75 mg of pollen and analysed using samples with ($n = 3$) and without ($n = 3$) pollen disruption. We also extracted lipids from dried egg powder as an external standard with a known amount of lipids. Dried pollen was first treated with cold acetone (750 µl in a 1.5 ml vial) to remove the external pollen coat. The vials were then placed in a water bath above the boiling point of acetone (56°C) to evaporate the solvent. The lipids that remained were resolubilized using 600 µl of a 2 : 1 chloroform : methanol ($CHCl_3$: MeOH) mixture with phase separation [55]. We used $CHCl_3$: MeOH because of its extraction efficiency with respect to different lipid classes [60]; $CHCl_3$ is efficient at extracting

nonpolar lipid fractions, while MeOH extracts polar lipids. After the fractions were separated, the 2 : 1 $CHCl_3$: MeOH wash was applied two more times to capture any remaining lipids in the vial. At the end of the process, the combined mass of the extracted lipid fractions represented the total amount of lipids extracted from a sample.

Loveridge method—We used the same Folch extraction detailed above when performing the Loveridge method, but only removed the lipid fraction of each sample. The samples were then placed in a drying oven at 40°C until a constant weight was obtained. The mass of the delipidated dried pollen sample was subtracted from the starting mass of the dried pollen sample. The difference in mass from the starting mass of the pollen sample represented the amount of lipids extracted through this method [56].

Vanillin assay—For the vanillin assay, pollen was analysed in quadruplicates for lipid content using soya bean oil as the external standard, as described by two previous studies [73,74]. Briefly, 1 mg of pollen was washed with sodium sulfate to remove non-lipid contaminants. Pollen lipids were then extracted using three $CHCl_3$: MeOH washes. Each sample reacted with the vanillin reagent and was run on an Epoch Biotek microplate spectrophotometer against a soya bean oil standard at 525 nm. The lipid content was determined using a standard curve calculated to measure the amounts of unsaturated fatty acids in a sample.

(d) Fatty acid analysis

We identified and quantified the fatty acids of intact and disrupted *Brassica* and *Rosa* pollen via fatty acid methyl ester (FAME) analysis using gas chromatography and GC-MS. A 30 µg quantity of tetracosanoic acid was added to each sample as a quantitative internal standard and to ensure that FAME generation was complete. Samples were individually resuspended, pulverized and extracted three times in a 1 : 1 $CHCl_3$: MeOH mixture for 24 h. The $CHCl_3$: MeOH solution was removed, and the residue combined for each sample, evaporated and resuspended in MeOH : water (7 : 3) using three 1 min bouts of sonication. This solution was then extracted three times with an equal volume of water-equilibrated hexane. The hexane fractions were pooled, dried and saponified (the process of freeing fatty acids and glycerol from triglycerides) according to the procedures described by Ichihara & Fukubayashi [75]. The free fatty acids and the fatty acids freed upon saponification were converted into fatty acid methyl esters [75].

Identification of fatty acid methyl esters by gas chromatography with flame-ionization detector (GC-fid) was based on their retention times relative to standards on a DB-17 column (Agilent Technologies; this column had dimensions of 30 m length and 0.25 mm diameter, and 0.25 µm film thickness), using an Agilent 6890 Networked GC-fid outfitted with a 7683B auto-sampler. The system had a carrier gas flow rate of 1.2 ml min^{-1}, an inlet temperature of 280°C, a detector temperature of 290°C, and an oven ramp beginning at 50°C ascending at 5°C min^{-1} to a final temperature of 290°C, holding the final temperature for 20 min. The elution pattern of the fatty acid methyl esters on the GC-fid was confirmed by GC-MS (Agilent 5973), running the identical column, gas and temperature protocols as those described for the GC-fid. The mass of individual fatty acid methyl esters was calculated using area units under curves for identified compounds and previously generated standard curves. The full details of our FAME protocol can be found in electronic supplementary material, S6.

(e) Carbohydrate analysis

Intact *Brassica* and *Rosa* pollen samples were analysed for total carbohydrates using the phenol–sulfuric acid assay [76,77]. This colorimetric assay isolates soluble sugars, starch and fructosan [14]; these are the carbohydrate classes that are nutritionally available to most insects. This assay is a fast and efficient method that quantifies digestible plant saccharides but not cellulose, which is indigestible for most insects. The phenol–sulfuric acid assay measures samples at a wavelength of 490 nm. The full details of our digestible carbohydrate protocol can be found in electronic supplementary material, S7.

(f) Elemental analysis

Intact *Brassica* and *Rosa* pollen samples were analysed for a suite of ten elements: phosphorus (P), potassium (K), sulfur (S), calcium (Ca), magnesium (Mg), sodium (Na), zinc (Zn), iron (Fe), manganese (Mn) and copper (Cu). All samples were transferred to polypropylene digestion tubes and then digested using trace metal nitric acid on a 105°C graphite block. Following digestion, samples were brought to volume and analysed using Spectro axial CIROS inductively coupled plasma–atomic emission spectrometry [78,79]. The full details of our non-nitrogen elemental analysis protocol can be found in electronic supplementary material, S8.

(g) Statistical analysis

We used principal component analysis (PCA) to determine the effect of pollen disruption on the fatty acid profiles of *Brassica* and *Rosa* pollen. We used a two-way ANOVA test to compare the effects of pollen disruption on the total amount and type of fatty acids extracted from each pollen type. We also used a PCA to: (1) compare the collective elemental profiles, and (2) determine the effect of pollen disruption on the fatty acid profiles of *Brassica* and *Rosa* pollen. After checking for normality, we also performed a two-way ANOVA or Mann–Whitney–Wilcoxon test (depending on data normality) to compare differences between fatty acids from intact and disrupted pollen. Finally, *t*-tests were performed to compare differences between individual elements in *Brassica* and *Rosa* pollen.

4. Results

(a) Pollen disruption

Following the lipid removal of the *Brassica* pollen exine and its subsequent rehydration, we sampled a subset of pollen grains over a series of successive 30 s pollenkitt disruption time intervals. At each time interval we recorded the percentage of *Brassica* pollen grains that were disrupted. At 210 s (equal to seven 30 s disruption bouts) 90% of the *Brassica* pollen grains were disrupted, but for *Rosa* only 40% of the pollen grains were disrupted (figure 1*a*). Images of intact and disrupted *Brassica* pollen grains can be seen in figure 1*b*,*c*, respectively.

(b) Protein estimations

The Bradford assay on intact pollen yielded protein values of 8.9 ± 0.7% for *Brassica* and 17.4 ± 1.0% protein for *Rosa* (table 1). By contrast, the Bradford assay on disrupted pollen yielded protein values of 31.3 ± 0.7% for *Brassica* and 27.3 ± 1.4% for *Rosa*. This amounted to an increase in protein content of 244% for *Brassica* pollen and 59% for *Rosa* pollen, compared with non-disrupted *Brassica* and *Rosa* pollen, respectively. The Dumas combustion analysis yielded nitrogen values of 4.70 ± 0.01% for *Brassica* and 3.08 ± 0.01% for *Rosa*. Using the standard conversion factor of 6.25, this

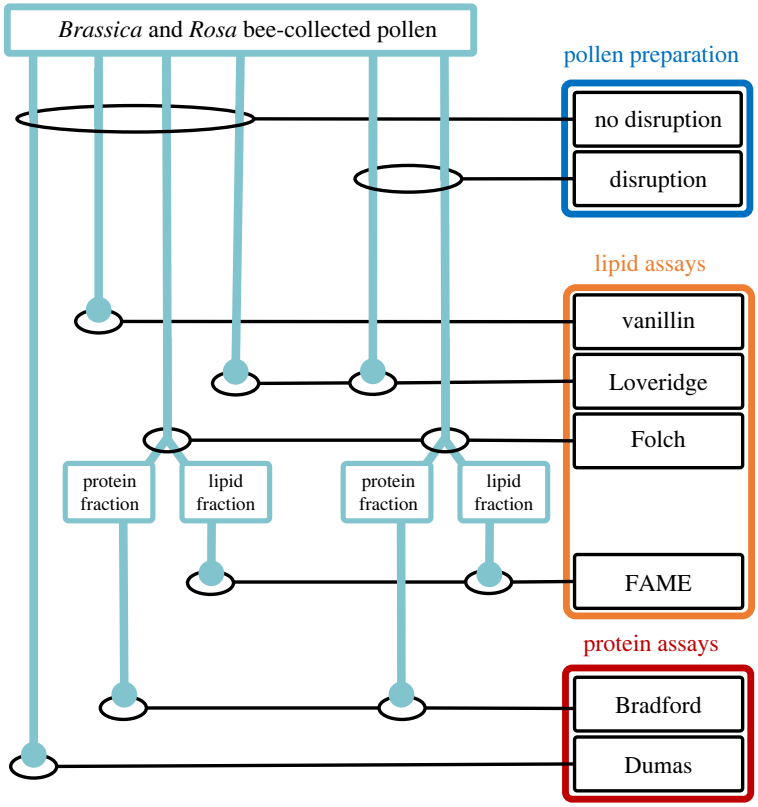


Figure 1. Pollen preparation flowchart for protein and lipid assays performed on *Brassica* and *Rosa* pollen. The effects of pollen disruption on pollen nutrient quantification were assessed for one protein assay (Bradford) and three lipid assays (Loveridge, Folch and fatty acid methyl ester (FAME)). The Dumas is a combustion assay that is completely destructive, so pollen disruption does not need to be performed prior to the analysis. Each vertical line coming down from the top box represents a set of two 75 mg samples of *Brassica* and *Rosa* pollen. Lines passing through or ending up in a black lasso indicate that the corresponding pollen sample has received the treatment shown on the right-hand side. (Online version in colour.)

Table 1. Influence of pollen grain disruption on protein content assessment. Water-soluble protein content (mean% ± s.e.) of *Brassica* and *Rosa* pollen estimated using the Bradford assay. Separate analyses were performed on non-disrupted and disrupted pollen.

	protein content	
pollen disruption	*Brassica*	*Rosa*
intact	8.9 ± 0.7	17.4 ± 1.0
disrupted	31.3 ± 0.7	27.3 ± 1.4

translated to pollen protein levels of 29% for *Brassica* and 19% for *Rosa* (table 1).

(c) Lipid estimations

The Folch assay on intact pollen generated lipid values of 19.1 ± 1.6% for *Brassica* and 20.6 ± 0.5% for *Rosa* (table 2). The values for disrupted pollen were 24.1 ± 0.1% for *Brassica* and 19.1 ± 1.8% for *Rosa*. The disruption process amounted to an increase in lipid content of 26.2% for *Brassica* pollen, and a decrease of 7.3% for *Rosa* pollen. We recorded lipid content of 42.0 ± 0.6% for our external standard (dried egg powder). The Loveridge assay on intact pollen generated lipid values of 20.2 ± 1.6% for *Brassica* and 21.1 ± 1.0% for *Rosa*. The values for disrupted pollen were 25.1 ± 1.0% for *Brassica* and 19.5 ± 2.1% for *Rosa*. In this case, the disruption process resulted in an increase in lipid content of 24.3% for *Brassica* pollen and a decrease of 7.6% for *Rosa*. The Loveridge assay returned a lipid content of 47.0 ± 1.0% for the dried egg powder control. Finally, the vanillin assay (using intact pollen) generated values of 11.1 ± 1.4% lipid for *Brassica* and 9.6 ± 0.9% for *Rosa* (figure 2).

Table 2. Influence of the analytical method and pollen grain disruption on the assessment of pollen lipid content. Total lipid content (mean% ± s.e.) of *Brassica* and *Rosa* pollen measured using three different assays (Folch, Loveridge and vanillin) with either non-disrupted or disrupted pollen.

	lipid content	
analytical method	*Brassica*	*Rosa*
Folch assay		
intact	19.1 ± 1.6	20.6 ± 0.5
disrupted	24.1 ± 0.1	19.1 ± 1.8
Loveridge assay		
intact	20.2 ± 1.6	21.1 ± 1.0
disrupted	25.1 ± 1.0	19.5 ± 2.1
vanillin assay		
intact	11.1 ± 1.4	9.6 ± 0.9

(d) Fatty acid identification and quantification

We identified seven fatty acids in *Brassica* pollen. Three of these were saturated (myristic (14 : 0), palmitic (16 : 0) and

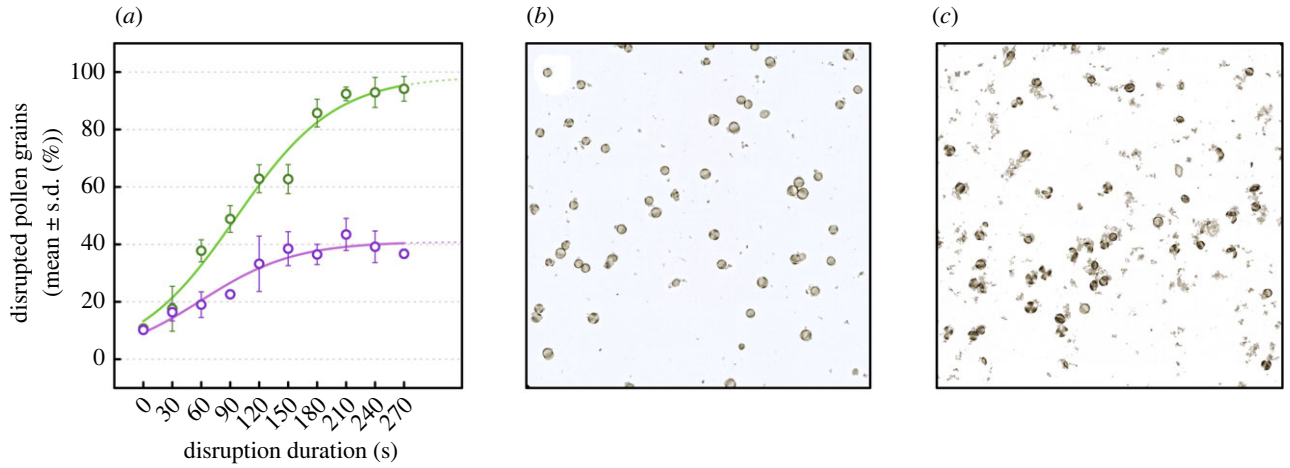


Figure 2. Determining the duration of the disruption step using *Brassica* pollen. (*a*) Proportion of pollen grains disrupted as a function of the duration of the mechanical disruption for *Brassica* (green, top line) and *Rosa* pollen (purple, bottom line). Each value represents the mean percentage (±s.e.) of disrupted pollen grains in three 1 × 1 mm squares after successive 30 s bouts of mechanical disruption. The logistic model fitted to the data (solid curves) helps to visualize the maximum proportion of disrupted pollen reachable with mechanical disruption. (*b*) Non-disrupted *Brassica* pollen grains. Picture of a 1 × 1 mm square with *Brassica* pollen grains before mechanical disruption. (*c*) Disrupted *Brassica* pollen grains. Picture of a 1 × 1 mm square with *Brassica* pollen grains after nine 30 s bouts of mechanical disruption. (Online version in colour.)

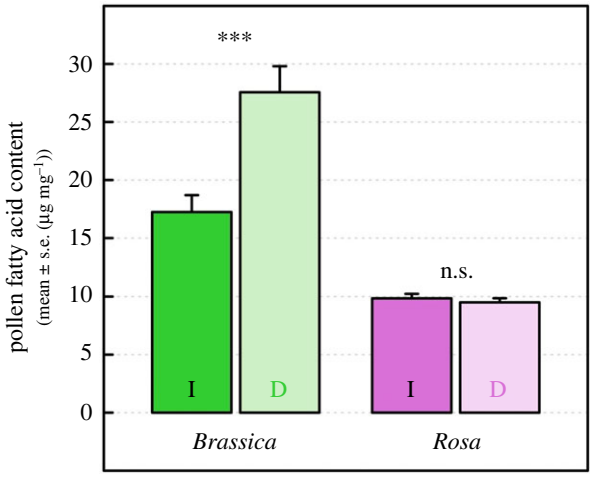


Figure 3. Influence of pollen grain disruption on the total amount of fatty acid extracted from *Brassica* and *Rosa* pollen. Mean mass of fatty acids extracted per mg (±s.e.) of *Brassica* (green bars) and *Rosa* (purple bars) pollen is reported for intact grains (I; darker bars) and disrupted grains (D; lighter bars). The total fatty acid content was calculated by the addition of the amounts of myristic acid, palmitic acid, roughanic acid, stearic acid, oleic acid, linoleic acid and α-linolenic acid detected by GC-MS. The annotations over the bars represent the significance of the difference in fatty acid content between intact and disrupted pollen grains (with *** corresponding to $p < 0.001$ and n.s. to $p > 0.05$).

stearic (16 : 0)), while the other four were unsaturated (roughanic (16 : 3), oleic (18 : 1), linoleic (18 : 2) and α-linolenic (18 : 3)). We identified five fatty acids in *Rosa* pollen (palmitic, stearic, oleic, linoleic and α-linolenic). The total fatty acid amount was significantly higher in disrupted compared with intact *Brassica* pollen (Mann–Whitney–Wilcoxon test: $Z = -2.8$, $p = 0.005$; figure 3). By contrast, there was no difference in the total fatty acid concentration between intact and disrupted *Rosa* pollen (Mann–Whitney–Wilcoxon test: $Z = 0.51$, $p = 0.610$; figure 3).

The PCA on the fatty acid profiles from intact and disrupted pollen grains showed two distinct clusters for *Brassica* pollen (figure 4*a*). The most influential variables from *Brassica*'s component 1 (56.8%) were (in descending order): (1) α-linolenic acid (two-sample *t*-test: $t = -3.9$, d.f. = 12, $p = 0.002$), (2) oleic acid (two-sample *t*-test: $t = -1.9$, d.f. = 12, $p = 0.11$), and (3) palmitic acid (two-sample *t*-test: $t = -5.5$, d.f. = 12, $p < 0.001$) (figure 4*b*). Fatty acids in *Rosa* were not significantly different between the intact and disrupted treatment groups (figure 5), with no clear separation between the intact and disrupted *Rosa* pollen. The most influential variables in component 1 (53.6%), despite there not being a significant effect of disruption on *Rosa* pollen, were (in descending order): (1) stearic acid (two-sample *t*-test: $t = 1.77$, d.f. = 12, $p = 0.10$), (2) oleic acid (Mann–Whitney–Wilcoxon test: $Z = 0.64$, $p = 0.52$), (3) linoleic acid (two-sample *t*-test: $t = -0.66$, d.f. = 12, $p = 0.52$), and (4) palmitic acid (two-sample *t*-test: $t = -1.7$, d.f. = 12, $p = 0.10$).

(e) Carbohydrate analysis

The phenol–sulfuric acid assay on intact pollen yielded digestible carbohydrate values of 33.0 ± 3.0% for *Brassica* and 33.0 ± 2.0% for *Rosa*. These values were not significantly different from one another (Mann–Whitney–Wilcoxon test: $Z = 0.00$, $p = 1.00$).

(f) Elemental analysis

The PCA revealed significant differences in the elemental composition of *Brassica* and *Rosa* pollen. Each element was an influential variable in the first component (95.5% of the variation), suggesting that the relative amounts of elements between the two pollen types were significantly different from each other (figure 5*a*). Component 2 explained 3.2% of the variation, with Na and Ca as the influential variables. There were distinct clusters of the elemental composition between *Brassica* and *Rosa* pollen (figure 5*b*). When individual elements were compared, we detected significantly higher levels of P (1.8×), S (1.7×), K (1.5×), Ca (1.4×) and Mg (1.2×) in *Brassica* pollen compared with *Rosa* pollen (table 3). By contrast, we detected significantly higher levels

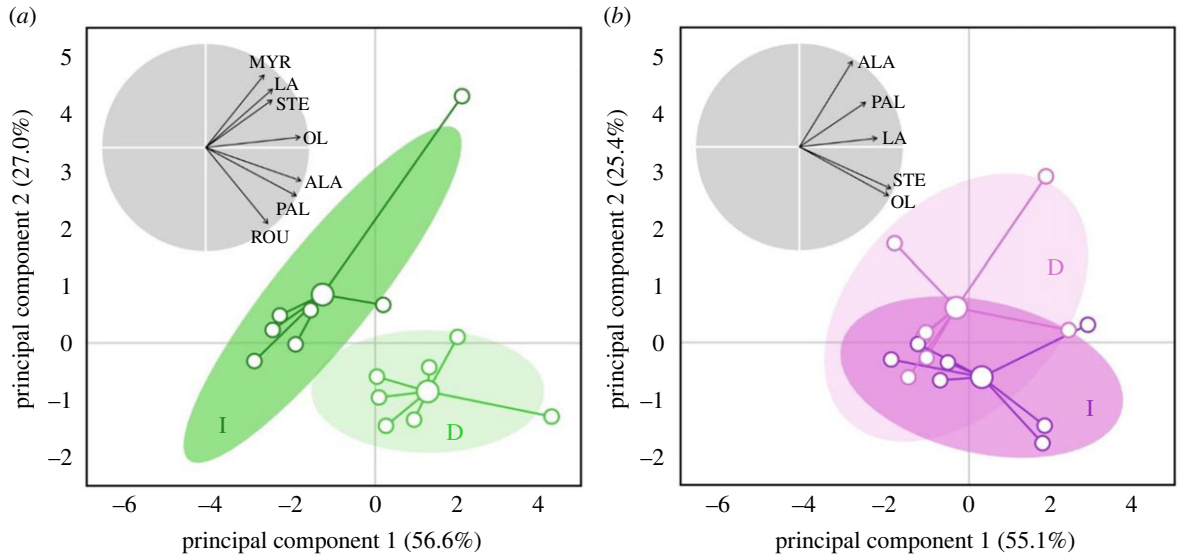


Figure 4. Representation of the influence of disrupting pollen grains on the fatty acid profiles of *Brassica* and *Rosa* pollen. Principal component analysis (PCA) of the main fatty acids detected by GC-MS in *Brassica* (*a*) and *Rosa* (*b*) pollen. The ellipses correspond to the 75% confidence intervals for the intact (I; darker colour) and the disrupted (D, lighter colour) data points. The two-dimensional representations explain 83.6 and 80.5% of the total variance for *Brassica* and *Rosa*, respectively. The correlation circles (in grey, top-left corners) represent correlations between the different fatty acids detected by GC-MS. These included: myristic acid (MYR), stearic acid (STE), palmitic acid (PAL), roughanic acid (ROU), oleic acid (OL), linoleic acid (LA) and α-linolenic acid (ALA). MYR and ROU were not detected in *Rosa* pollen. (Online version in colour.)

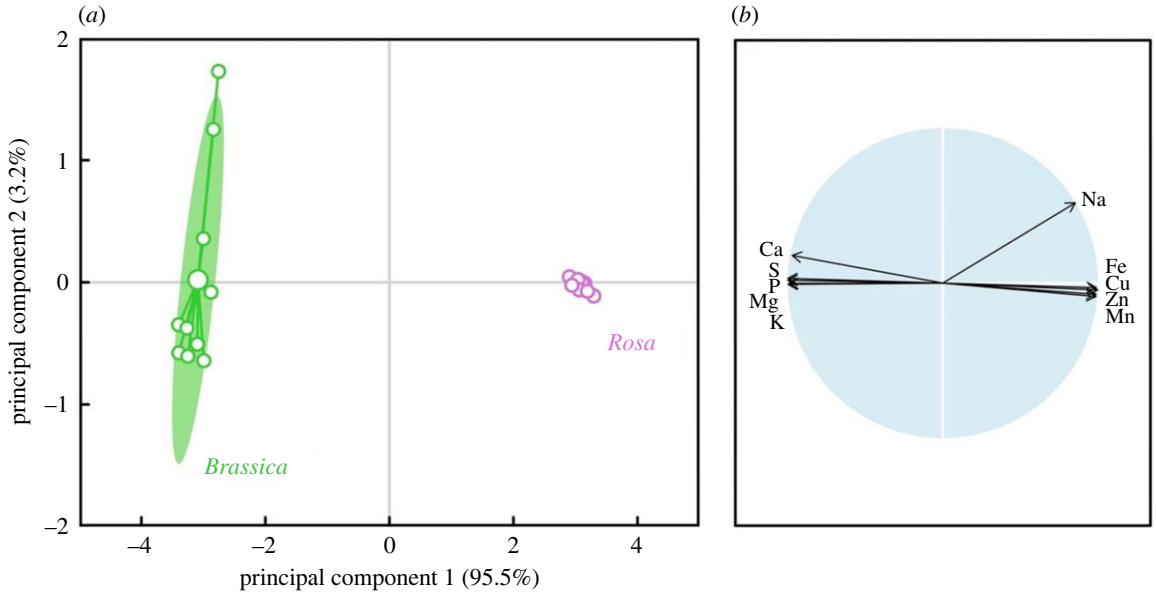


Figure 5. Representation of the elemental composition of *Brassica* and *Rosa* pollen. (*a*) principal component analysis (PCA) on ten elements (P, K, S, Ca, Mg, Na, Zn, Fe, Mn and Cu) for *Brassica* (green) and *Rosa* (purple) pollen with 75% confidence interval ellipses. The two-dimensional representation explains 98.8% of the total variance. (*b*) Correlation circle representing the nature of the correlations between elements. (Online version in colour.)

of Na (1.1×), Zn (1.8×), Fe (3×) and Cu (2.1×) in *Rosa* pollen compared with *Brassica* pollen (table 3).

5. Discussion/conclusion

It is now well established that significant variation exists in the amounts of individual nutrients contained within different pollen types [5]; protein ranges from 10 to 60%, lipids from 2 to 20% and carbohydrates from 1 to 40%. However, as we document in the current study, a significant proportion of this variation could likely be attributed to the use of different methods to estimate any single nutrient class. Progress in our understanding of the nutritional ecology of bees and other pollen feeders would thus benefit from a unifying approach for characterizing the multidimensional nutritional content of pollen. This is essential to allow comparisons among different studies. Based on our results, we make the following recommendations. First, pollenkitt disruption should be performed prior to conducting any non-destructive analytical method (e.g. gravimetric, colorimetric, GC-MS, HPLC). Second, with respect to estimating pollen protein content, the Dumas method is preferred. Third, from a nutritional perspective, fatty acid profiles provide a more accurate representation of pollen nutritional value compared with total lipid content. Fourth, a wet digestion method for plant tissues is a simple and powerful approach for characterizing a broad suite of micronutrient-related elements linked to animal physiological processes. Finally, although not

Table 3. Pollen element analysis. Summary of *Brassica* and *Rosa* pollen spectrometric assays of a suite of ten elements. Values for each element are reported in ppm (mean ± s.e.). Grey cells indicate that the concentration of the element is significantly higher in this pollen type compared with the other, using a *t*-test ($p < 0.001$).

	concentration (ppm)	
element	*Brassica*	*Rosa*
phosphorus	7205.2 ± 35.1	3953.7 ± 16.0
potassium	6217.4 ± 44.3	4099.3 ± 16.0
sulfur	3683.3 ± 17.1	2117.5 ± 5.4
calcium	2536.3 ± 36.8	1795.7 ± 18.9
magnesium	1576.5 ± 8.2	1280.7 ± 6.0
sodium	85.8 ± 1.4	94.9 ± 0.0
zinc	42.7 ± 1.0	76.7 ± 0.9
iron	35.0 ± 1.0	104.0 ± 2.6
manganese	25.0 ± 0.1	32.0 ± 0.2
copper	9.5 ± 0.1	19.1 ± 0.1

examined in this study, sterol content should be measured in pollen, as bees and other pollen-feeding arthropods have a dietary requirement for sterols. Below we provide our rationale for making these suggestions.

Traditionally, most analytical techniques used in pollen research treat pollen grains like plant vegetative tissues. However, because of the unique physical properties of pollen, especially the pollenkitt, proper pollen nutrient analysis requires additional steps. In particular, pollen disruption benefits any analytical technique that requires a nutrient extraction step because it often improves, quantitatively and qualitatively, extraction yield [28,29,80]. Interestingly, pollen shares physical and chemical characteristics with algae, an organism that is extensively studied for biofuel research. Akin to pollen grains, algae cells are small and possess a thick and complex wall [81,82]. This required researchers to develop techniques to disrupt the algal cell wall to release intact lipids and maximize extraction efficiency [81–83]. When we disrupted *Brassica* and *Rosa* pollen grains, we observed two key results. First, the effect of pollen disruption varied depending on pollen type—90% of Brassica pollen grains were disrupted after 210 s of homogenization, but after 270 s of homogenization only 40% of *Rosa* pollen grains were disrupted. This finding is similar to those by Swenson & Gemeinholzer [80], who reported extreme variation in pollen wall disruption owing to species-specific variation in pollen structure, which can vary in shape, size and durability [25,84–86]. Other methods to disrupt pollen, including vigorous overnight vortexing or ultrasonication [28,87,88] and different buffers [89], have also been used, but it is not known how effective they are among different pollen taxa. Second, comparisons of total lipid or total fatty acid content between intact and disrupted *Rosa* pollen showed no significant difference. This suggests that while some pollenkitt types may be difficult to disrupt (i.e. that of *Rosa* pollen), this kind of toughness is not necessarily correlated with porosity (solvent permeability). However, as our results with *Brassica* pollen suggest, the benefits of disrupting the pollenkitt (i.e. increased extraction efficiency) seem to outweigh the costs (i.e. time spent disrupting pollen).

Historically, studies of the nutritional value of pollen for bees were primarily focused on protein levels because there is a strong correlation between an insect's performance and the amount of protein in its diet [13,35,90]. In the current study we used two of the most common approaches for measuring pollen protein levels: the Bradford assay (colorimetric method that uses soluble protein content to estimate protein levels) and the Dumas assay (N-based method that uses a conversion factor, typically 6.25) to estimate protein levels from nitrogen content). Using the Bradford assay, we observed higher protein levels for disrupted versus non-disrupted pollen (3.4× higher for *Brassica*, and 1.6× higher for *Rosa*). Our findings are consistent with other studies [28,91], which reported similar increases in protein estimations following disruption of the pollenkitt. Given that proteins occur in different regions of pollen grains, but a significant proportion are contained inside the pollenkitt [92], a low protein estimation for non-disrupted pollen (especially for *Brassica*) suggests that an intact pollenkitt can greatly reduce the utility of the Bradford assay. The Dumas assay, in contrast, is a destructive sampling method that captures all the nitrogen contained outside and inside a pollen grain. We reported N values of 4.7% for *Brassica* and 3.0% for *Rosa*, which converted to 29 and 19% protein, respectively (using a 6.25 conversion factor). Theoretically, the Bradford assay results should not exceed these values because the Dumas already overestimates protein content by including non-proteinaceous nitrogen and the 6.25 conversion factor is higher than suggested conversion factors for pollen (e.g. 5.6 [38]). However, the Bradford protein values using disrupted pollen were higher for both pollen types. This suggests that the Bradford assay largely overestimates protein content in disrupted pollen (7.2% for *Brassica*, and 42% for *Rosa*), with strong variation depending on pollen taxa. Potential causes for this overestimation were outlined above in the 'Review of methods' section (see §2b(ii)).

There is growing evidence that besides proteins, lipids are an important macronutrient for animals, including bees [5]. For example, lipids influence nutrient regulation behaviour of bumblebees [71], and can affect learning performance in honeybees [93]. To assess any differences among methodologies for measuring pollen lipid content, we first compared three methods commonly used to quantify total lipids in non-disrupted pollen. While we recorded similar values for the Folch and Loveridge assays, the vanillin assay yielded values lower than half those reported for the Folch and Loveridge assays. This likely reflects the fact that the vanillin assay solely measures unsaturated fatty acid content [61,62]. This also aligns with our fatty acid analysis, which detected a mixture of saturated and unsaturated fatty acids in both *Brassica* and *Rosa* pollen. These results provide strong evidence that the vanillin assay is a poor choice for researchers interested in quantifying total lipid content. With disrupted pollen, the Folch and Loveridge assays returned significantly higher values compared with non-disrupted pollen. In addition, and similarly to what was observed with the Bradford assay, the increased lipid values were observed following pollenkitt disruption in *Brassica* pollen, but not *Rosa* pollen. This suggests the pollenkitt of *Rosa* may be more permeable to solvents than that of *Brassica*. Finally, concerning the dried egg powder control (which has a reported lipid content of 38%), our results suggest that the Folch assay is more accurate than the Loveridge assay when estimating total lipid levels in pollen.

The total lipid pool in pollen consists of a broad class of lipids, but only a subset (fatty acids) has been directly linked to bee health. Thus, a focus on the total lipid content of pollen types may not be particularly informative. Recent work with honeybees has demonstrated that fatty acids have substantial nutritional value, and that the quantity and quality of fatty acids available for consumption affect pollinator performance [94] and learning [93,95]. We found very distinct fatty acid profiles for *Brassica* and *Rosa* pollen and observed that the total fatty acid content of *Brassica* was higher compared with *Rosa*. We also showed that pollenkitt disruption affects the fatty acid yield quantitatively and qualitatively, especially in *Brassica* pollen. This is in alignment with recent work by Wu *et al*. [29]. However, disrupting *Rosa* pollen did not influence quantitative or qualitative fatty acid yield. This information, together with the total lipid data, suggests that pollen from different plant species responds differently to pollenkitt disruption. Although some pollen types may not require disruption, we recommend doing so as a general practice, as it does not appear to negatively impact the results for lipid analyses. Additionally, given the variability among pollen types in pollen coat toughness and permeability, disrupting the pollenkitt prior to total lipid analysis ensures consistency across studies and increases the reliability of the lipid data being generated.

The third pollen macronutrient class is digestible carbohydrates (e.g. monosaccharides, disaccharides and starch). We observed no significant difference in the digestible carbohydrate content of *Brassica* and *Rosa* pollen collected by honeybees; both registered at 33% dry mass. The majority of pollen nutritional studies have used bee-collected pollen as their pollen source [96], and it is estimated that *ca* 40% of the digestible carbohydrates in bee-collected pollen comes from nectar or honey applied by the bee foragers [5]. This application serves a functional purpose: it binds the pollen together for transportation back to the hive. It also likely helps preserve collected pollen [97]. There is little current data to suggest that digestible carbohydrate levels in pollen influence foraging behaviour, given that honeybees primarily meet their energy needs through a combination of honey and nectar [11].

Pollen also contains a full suite of micronutrients important for pollinator health. These can be present as biomolecules (which can be arduous to identify and quantify) and as free ions (e.g. PO_4^{3-}, Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, Fe^{2+}, Zn^{2+} and Mn^{2+}). A relatively straightforward approach to measure these micronutrients is at the elemental level [18]. We currently know little about which nutrient-linked elements in pollen—especially those that serve important physiological and maintenance functions (e.g. Na, K, Ca, Mg), or structural purposes (e.g. Fe, Zn, Mn)—might affect bee health. However, new insights are beginning to emerge, for both wild bees [98–100] and honeybees [101]. Importantly, two aspects make an elemental approach particularly attractive. First, it is thorough; samples are completely digested so there is no need to include a pollenkit disruption step. Second, a single sample generates values for 10 or more unique elements. As shown in table 3, we recorded high concentrations of: (1) phosphorus (P), linked to nucleic acids and energetic nucleotides; (2) sulfur (S), a key element in sulfur-containing amino acids (e.g. methionine and cysteine) linked to longevity and overall performance [100]; and (3) potassium (K), calcium (Ca) and magnesium (Mg), which are all linked to important physiological and maintenance functions. Interestingly, sodium (Na) was recorded at generally low levels, which raises interesting questions about how bees meet Na requirements given the importance of Na in nervous system function [102,103]. Perhaps this explains the observation that honeybees drink 'dirty water' as a way to supplement their Na intake [103–105]. The metals zinc (Zn), iron (Fe) and copper (Cu), which are all linked to structural function, were found at generally low concentrations (table 3). Given the ability to link elements with host plants and their consumers [79], there is value in linking pollen availability at the landscape level with pollinator diversity and abundance. Filipiak *et al*. [106] recently showed how micronutrient deficiencies in pollen, specifically P, K, Cu and Zn, can negatively impact performance of the wild bee *Osmia bicornis*. They suggest that bee conservation measures would benefit from focusing on the micronutrient composition of available pollen. Specifically, they encourage selection of plants that provide pollen that is nutritionally complementary in multiple dimensions for bees.

Sterols, a class of amphipathic lipids, are the final pollen micronutrient worth discussing even though we did not analyse them in the current study. Like all insects, bees use sterols as important cell membrane inserts and as precursors to the hormones that regulate moulting [107]. However, all arthropods lack the biosynthetic ability to make sterols de novo and thus require a dietary source [17]. The sterols that have been recovered from pollen are diverse, but 24-methylene cholesterol, β-sitosterol and isofucosterol are generally the most common and abundant [88,108]. Recently, Zu *et al*. [109] conducted a broad survey of pollen sterols (total amounts and composition) using 122 plant species (105 genera, 51 families). They recovered 25 different pollen sterols and found the sterol profiles to be associated with phylogeny and environment (e.g. temperature, seasonality, precipitation). Interestingly, dietary sterol profile can have important negative ramifications for insects, especially for generalist feeders that mix their diets and consume significant amounts of sterols that are structurally suboptimal for cell membrane dynamics or as a precursor for moulting hormone [110]. Thus, there is value in documenting total sterol amounts and profiles in pollen. Like fatty acid methyl ester (FAME) analysis, sterol analysis is performed using GC-MS. As this process requires solvent-based extraction, we recommend performing a pollenkitt disruption step prior to the initial extraction step.

A brief discussion is also warranted about pollen transport behaviour in bees, and its implication for pollen nutritional analyses. Bees collect pollen and return it to their nests, where it serves as a multidimensional nutritional resource for their offspring. In the case of eusocial bees, nurses consume pollen (in the form of bee bread) and then feed the developing larvae with mandibular and hypopharyngeal gland secretions [111]. While a small number of bee taxa transport pollen internally in their crop (the ancestral condition), most transport it on the outside of their body, typically on the ventral metasoma (most Megachiladae) or, more commonly, on the hindleg tibia [112]. Bees that pack pollen on their hindlegs transport it in one of three modes: moistened with nectar or oils, dry, or glazed (a combination of dry and moist) [113]. When bees moisten or glaze pollen, it necessarily modifies the nutritional composition of pollen as a resource. Thus, pollen source—either hand-collected or bee-collected—needs to be documented when pollen nutrient analyses are reported. Additionally, pollen source needs to be considered when results are being extended to bee nutritional ecology.

In conclusion, we believe a unifying approach with respect to pollen nutrient content can greatly inform and advance our understanding of bee nutritional ecology, and by extension bee health. We provide a framework for standardizing methods that will facilitate comparisons among numerous, independent pollen nutrient content studies. We recognize there are trade-offs for any method (including time, access to equipment, budget, etc) but we encourage researchers to employ our approach when they can. As we highlight in this study, a recurring issue with available protein and lipid assays is that they require some prior knowledge of pollen amino acid or fatty acid content. For instance, when analysing protein content using a colorimetric method (i.e. Bradford), results depend on the amino acid profile (the dye used reacts with basic and aromatic amino acids). Likewise, the result of the vanillin assay (a colorimetric lipid assay) reflects the fatty acid composition of pollen as the reagent only reacts with saturated fatty acids. The acquisition of detailed information on pollen amino acid and fatty acid profiles demands time-consuming experiments and expensive analytical equipment. However, the development of a collaborative pollen nutrient content database, focused on pollen amino acid content and fatty acid profiles, would alleviate some of these issues. We champion such an approach as it would provide great support in the effort to unify research about the nutritional ecology of bees, as well as that of other pollenfeeders.

Data accessibility. Raw data are available on USDA Ag Data Commons: https://data.nal.usda.gov/e.

Authors' contributions. P.La: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, resources, writing—original draft, writing—review and editing; P.Le.: conceptualization, data curation, formal analysis, investigation, methodology, writing—review and editing; R.J.G.: data curation, formal analysis, investigation, methodology, writing—review and editing; J.R.: conceptualization, funding acquisition, resources, writing—review and editing; S.T.B.: conceptualization, methodology, resources, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed herein.

Conflict of interest declaration. The authors declare that they have no competing financial interests.

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